
Original Article

GICS or ICB, how different is similar?

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ABSTRACT A number of different classification systems exist to assist portfolio managers in diversifying exposure to industry- or sector-specific risk. Recently a new system was added, the Industry Classification Benchmark (ICB). Both this framework and the Global Industry Classification System (GICS) are very popular among practitioners. The two frameworks are reported to be similar in methodology and equivalent in use. This article therefore assesses three questions concerning these two systems: the homogeneity of the classes, their relation to market returns and the stability of the sectors for variations in the market. The results indicate that differences do exist between the two systems. These differences, however, are concentrated around individual sectors and are the result of classification decisions inherent in the methodology. For large-cap companies, such as the sample used in this study, the impact of these differences is relatively limited, which does not guarantee the same result for mid- and small-cap stocks. The two methodologies are therefore globally similar and locally different.

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INTRODUCTION

Portfolio managers are often faced with the challenge of reducing their exposure to equity sector-specific risk. They are aided in their task by several classification frameworks grouping equity into economically relevant sectors. Recently a new framework joined the list, the Industry Classification Benchmark (ICB). ICB resembles closely, both in structure and methodology, an already existing classification framework, the Global Industry Classification System, (GICS). GICS was jointly developed by

Standard and Poor's (S&P) and Morgan Stanley Capital International (MSCI). ICB is developed by Dow Jones and FTSE. Both in structure and economic validity, the two systems are remarkably similar. Developed and distributed by major index providers, they are widely available and used in the construction of various indexes leading to an ever-rising popularity, especially among practitioners.

Similar, however, does not mean equal. A portfolio manager may quickly acknowledge that the economic validity of the two systems

is equivalent. A second series of questions then becomes important. First, how homogeneous are the constructed sectors from either of these systems. Second, is there any difference between the two systems in the dependence of returns upon the movements in the market, that is, the alphas and betas. Third, how stable are the constructed sectors when the market moves. This article looks into these three questions, thus comparing GICS and ICB from a practitioner's point of view.

Past research in this area usually focused mostly on either the economic validity of the different sector classification systems or the importance of sector effects compared to, for instance, country, region or style effects. The classification method and considered period tend to influence the results greatly. Beckers *et al* (1996), Heston and Rouwenhorst (1995) and L'Her *et al* (2002) consider the period between 1975 and 2000. They conclude that country specific risk is of higher importance than the risk specific to industry allocation. Baca *et al* (2000), Cavaglia *et al* (2000) and Hamelink *et al* (2001) suggest the opposite and conclude that country-specific risk overshadows industry-specific risk. This view is confirmed by Beine *et al* (2006) who argue that contagious effects are less present in portfolios diversified according to industry rather than country. Clarke (1989) subsequently questioned the validity of sector classification if the production method, end product or product market hardly impacted the actual classification of equity. Statistical decomposition methods and clustering techniques (Elton and Gruber, 1970; Farrel, 1974; Brown and Goetzmann, 1997; Kao and Schumaker, 1999) suggested therefore much more flexible classification methods.

From the view point of a portfolio manager wanting to eliminate sector-specific risk, these questions are of secondary importance. The actual risk that could be reduced using either one of the suggested techniques is the prime objective. Bhojraj

et al (2003) and Chan *et al* (2007) compared the existing frameworks: Fama and French (1997), GICS, NAICS and SIC. These systems differ in the number of categories, the structure and criteria used to define sectors and the availability and purpose they are used for. Both studies concluded the GICS framework to be efficient in constructing homogeneous sectors, with relatively low correlation outside the sectors while using less categories than most counterparts. Chan *et al* (2007) find the GICS framework to be more effective in finding homogeneous sectors, compared to statistical clustering techniques.

The increasingly popular GICS framework is now joined by the ICB framework. As both are remarkably similar, the task of comparing these systems becomes interesting. Answering the three questions mentioned above shall, therefore, be the main objective of this article. In order to do so, a three-part methodology will be used. First, the homogeneity of the categories is compared for GICS and ICB. The approach suggested by Chan *et al* (2007) is used for this purpose. Second, using a Single Index Market Model (SIMM), the performance of the different sectors for both GICS and ICB is compared to a benchmark. Differences in the construction of the sectors that might effect the behavior of the sectors with respect to the selected benchmark can, in this case, be assessed. Third, using a Quadratic Market Model (QMM), it is possible to assess correlation shifts of the sectors when a shift occurs in the benchmark. Again, differences between the two systems can be assessed in this context.

The results are rather surprising. Although the similarity of the structure is remarkable, GICS seems to achieve higher homogeneity than its counterpart, using less categories. Eventually, the two systems will reach a similar level of homogeneity when the number of categories reaches its maximum. The differences are mostly concentrated around a limited number of

sectors in which the GICS framework has an advantage over ICB. The heterogeneity outside the clusters is, however, remarkably similar for the two systems. As the sectors show similar outside correlation, the performance and dependence upon the general evolution of the stock market will be fairly similar across all sectors. Similarly, shifts in correlation result in comparable results for both systems.

The remainder of this article will be structured as follows. The next section gives a concise overview of the two considered sectoring methodologies. The subsequent section presents the methodology used in the article. The data used for the analysis are presented in the fourth section. Results are discussed in the penultimate section 5 after which a conclusion can be drawn about the similarity of the two frameworks.

GICS AND ICB METHODOLOGIES

GICS was introduced in 1999 by MSCI and S&P. It is structured using a four-layer approach starting with 10 sectors, 24 industry groups, 67 industries and 147 sub-industries. The classification into each of these categories is based upon the market-oriented approach to sectoring rather than on production characteristics. In this case, the market responsible for most of the company's revenue determines which sector or industry a company belongs to. Secondary elements come into play to refine this classification, such as earnings analyses and market perception. Each year the company's classification is reviewed and also when a major event changes the company's primary business.

The ICB was jointly developed by Dow Jones and FTSE. Again, the classification consists of four layers: industry, super-sector, sector and sub-sector. The industry layer consists of 10 industries, similar to the GICS framework, followed by subdivisions in 19 super-sectors, 41 sectors and 114 sub-sectors. Companies are assigned to a sector based on

their major source of revenue. If more than one source is considered, the allocation to a sector is determined by an analysis of the audited accounts and the director's report. The companies can, therefore, be allocated to a sector either on the basis of the market in which they operate or the end use of the product they produce.

Even though the two systems are quite similar, there are some differences that should be noted. First, the consumer businesses are considered differently in the two cases. GICS divides them into cyclical and non-cyclical businesses, whereas ICB considers goods and services to be two distinct classes. In the second case, the market is of less concern and the type of product plays a bigger role. Second, some businesses are grouped differently in the two systems, again based on a possible distinction between product and market. Mining companies are part of the 'Basic Materials' sector in the case of ICB, whereas they are part of the 'Energy' in the case of GICS. These differences might be small, indeed; however, they reveal the difference in the classification philosophy used by the two systems. This small difference could well lead to more profound differences in sectoral homogeneity when looking at the performance of the sectors. A collapse in consumer confidence might well affect goods and services alike, but could result in less harm to cyclical goods rather than to non-cyclical or discretionary ones. Similarly, oil and gas prices are not necessarily affected by the same events as metals are.

METHODOLOGY

To answer the question whether the difference in methodology could already impair the homogeneity of the defined sectors, a two-part methodology is chosen. First, inside versus outside homogeneity of each of the sectors is quantified. For this, the methodology suggested by Chan *et al* (2007) is used. Second, it is necessary to assess whether any of the found differences have

repercussions for the dependence between sectors and the market. Chan *et al* (2007) computed the correlation inside a sector and compared this value to the correlation outside the sector. This step is repeated for each of the considered sectors across each of the four levels of the sector classification systems. Let K be the number of stocks in the data set. Using GICS and ICB, these stocks are classified according to their industry I . If industry I contains N stocks, we calculate the average pairwise correlation of the return of stock i within the industry as follows.

$$\rho_{iI} = \frac{\sum_{j \in I, j \neq i} \rho_{ij}}{N - 1} \quad (1)$$

where ρ_{ij} is the pairwise correlation between the returns of stocks i and j . Similarly, the correlation outside the industry is calculated using the following relation.

$$\phi_{iI} = \frac{\sum_{j \notin I} \rho_{ij}}{K - N} \quad (2)$$

We can now define the average inside and outside correlation across all the considered stocks.

$$\bar{\rho}_I = \frac{\sum_{i=1}^K \rho_{iI}}{K} \quad (3)$$

$$\bar{\phi}_I = \frac{\sum_{i=1}^K \phi_{iI}}{K} \quad (4)$$

The second part of the methodology consists of assessing whether the different industry classes behave differently in terms of correlation shifts, compared to a chosen benchmark. Sectors react differently to economic events depending on the nature of the events. The correlation between some sectors and the market in general, therefore, varies. Considering correlation to be a constant and proportional to market returns is therefore only interesting as the first step in the analysis. An SIMM suggested by Sharpe (1963) models returns of an asset as a linear function of the returns of the market.

$$r_{it} = \alpha + \beta_i r_{mt} + \varepsilon \quad t = 1, \dots, T \quad (5)$$

The actual evaluation of the correlation needs to be done using a different model. In order to model correlation shifts, we adopt the QMM proposed by Spurgin *et al* (2001). In this framework, the β is considered to be a linear function of the market returns.

$$\beta_{im} = a + b r_m \quad (6)$$

This assumption holds as long as the ratio of asset and market variances remains constant implying linear correlation. Since β_{im} is only the slope of the return function, the integral of equation (6) will give the relation for the return r_i of the asset in function of the market.

$$r_i = \int \beta_{im} dr = a r_m + \frac{1}{2} b r_m^2 + c \quad (7)$$

Spurgin *et al* (2001) argue that this equation can be estimated using Ordinary Least Squares Regression of the asset returns against the market returns.

$$\hat{r}_{it} = \hat{a} + \hat{b}_1 r_{mt} + \hat{b}_2 (r_{mt} - \bar{r}_m)^2 + \varepsilon \quad (8)$$

The excess return, beta and the correlation are function of a chosen benchmark, the stock market in our case. The excess return or $\hat{\alpha}$ of a particular sector can therefore be calculated by setting $\hat{b}_1$ equal to zero, as it represents the return independent from the market.

$$\hat{\alpha}(r_m) = \hat{a} + \hat{b}_2 (r_{mt} - \bar{r}_m)^2 \quad (9)$$

When the excess return equals the monthly average return, $\hat{\alpha}(r_m) = \hat{a}$. $\hat{\beta}(r_m)$ was set to depend linearly on the returns of the benchmark. Using the estimated regression parameters of equation (8), $\hat{\beta}(r_m)$ can be defined as follows:

$$\hat{\beta}(r_m) = \hat{b}_1(r_m) + \hat{b}_2(r_{mt} - \bar{r}_m)^2 \quad (10)$$

This definition quite simply leads to the definition of the correlation as it follows from.

$$\hat{\rho}_{im}(r_m) = \hat{\beta}(r_m) \frac{\sigma_s}{\sigma_b}$$

DATA¹

The data used for the present study are the components of the S&P 500 index for a period covering 20 years, from 1988 to 2007 included. The data are monthly closing prices for the index. From the monthly closing prices, continuous monthly returns are derived using the following relation,

$$r_{it} = \log(P_t) - \log(P_{t-1}) \quad (11)$$

where P_t is the closing price at time t . Several test samples are created using the considered index. First, a general sample covering the total period from 1988 to 2007 is constructed eliminating all companies, which do not cover the entire period. Second, to avoid stationary problems of the correlation structure of the considered stocks, over a long time period, four non-overlapping sub-samples are constructed of 5 years length each. Again, only the companies present in the entire sub-sample are taken into account. The GICS and ICB codes are used to classify the stocks. The choice of the S&P 500 sample is based on the fact that it is a good representation of the US stock market. Results found for this index can be seen as a good approximation of the results that would be found using larger indexes. The structure of the index implies the consideration of only large-cap companies, eliminating size effects in the results.

RESULTS

The results of the analysis are both interesting and remarkable. The results will be reported in two parts; first, the correlation analysis followed by the results of the SIMM and QMM estimations. In general terms, the findings indicate that the question of similarity of the two considered systems is not a closed one. The GICS framework seems to show more sectoral cohesion according to our findings for a number of sectors. This result should, however, be considered with some care and be compared with the findings of the second set of tests.

The SIMM and QMM models show results consistent with the postulate of similarity of the two classification systems. The sectors sensitive to correlation shifts demand a closer inspection of the found results.

The first part of the analysis consists of calculating the various inside- and outside-sector correlations for each of the four sub-samples constructed from the total data sample. The results of this analysis are reported in Tables 1–4. Inside-sector correlation is computed for each sector and for each of the levels in the classification systems. As the correlation coefficients reported in the table are all averages across the considered sector and level in the classification systems, effects of company-specific risk and outliers are greatly reduced. The conclusions that can be drawn from the presented data are rather interesting.

First, when looking at all the tested samples, one perceives a generally higher average inside-sector correlation for the GICS framework than for its counterpart, the ICB framework. This difference is present across all levels of the framework. The effects seem to trail off between the 6- and 8-digit level of the classification systems. One has to remember that the 4-, 6- and 8-digit levels of the GICS framework consist of more categories than those for the ICB system. However, in most cases, the inside-sector correlation for the GICS 4-digit level is equivalent to the 6-digit level for the ICB framework.

When looking at the level of outside-sector correlation, the findings are equally interesting. Contrary to the inside-sector findings, the differences between the two systems seem much less important. Across all levels of the classification frameworks, the average outside correlation is similar and declining for the four levels of each of the systems. The most important jumps in correlation occur between the 6- and 8-digit levels of the systems. The resulting effect is that a noticeable difference exists in terms of

Table 1: Inside versus outside correlation analysis for the two considered classification systems (period considered is 1988–1992)

<i>GICS sector</i>	<i>GICS code</i>	<i>GICS 2</i>	<i>GICS 4</i>	<i>GICS 6</i>	<i>GICS 8</i>	<i>ICB sector</i>	<i>ICB code</i>	<i>ICB 2</i>	<i>ICB 4</i>	<i>ICB 6</i>	<i>ICB 8</i>
<i>Within sector</i>						<i>Within sector</i>					
Energy	10	0.36	0.36	0.39	0.39	Oil & gas	500	0.33	0.33	0.34	0.37
Materials	15	0.38	0.38	0.38	0.20	Basic materials	1000	0.41	0.42	0.38	0.32
Industrials	20	0.41	0.42	0.43	0.42	Industrials	2000	0.38	0.38	0.43	0.46
Cons disc	25	0.34	0.37	0.36	0.28	Consumer services	5000	0.33	0.35	0.35	0.34
Cons stap	30	0.44	0.46	0.46	0.38	Consumer goods	3000	0.36	0.42	0.44	0.41
Health care	35	0.35	0.35	0.42	0.39	Health care	4000	0.36	0.36	0.37	0.43
Financials	40	0.36	0.40	0.40	0.37	Financials	8000	0.36	0.41	0.41	0.36
Info tech	45	0.37	0.37	0.34	0.36	Technology	9000	0.38	0.38	0.38	0.38
Telecom	50	0.60	0.60	0.60	0.60	Telecom	6000	0.35	0.35	0.35	0.35
Utilities	55	0.48	0.48	0.47	0.47	Utilities	7000	0.48	0.48	0.49	0.50
Simple mean		0.41	0.42	0.42	0.39	Simple mean		0.37	0.39	0.40	0.39
<i>Outside sector</i>						<i>Outside sector</i>					
Energy	10	0.15	0.15	0.15	0.11	Oil & gas	500	0.15	0.15	0.15	0.15
Materials	15	0.30	0.30	0.29	0.15	Basic materials	1000	0.32	0.32	0.29	0.23
Industrials	20	0.30	0.30	0.29	0.27	Industrials	2000	0.30	0.30	0.30	0.29
Cons disc	25	0.28	0.28	0.26	0.18	Consumer services	5000	0.29	0.29	0.29	0.25
Cons stap	30	0.31	0.31	0.30	0.24	Consumer goods	3000	0.30	0.30	0.30	0.25
Health care	35	0.28	0.28	0.28	0.27	Health care	4000	0.29	0.29	0.29	0.29
Financials	40	0.29	0.29	0.28	0.25	Financials	8000	0.29	0.29	0.29	0.26
Info tech	45	0.28	0.28	0.25	0.24	Technology	9000	0.28	0.28	0.28	0.28
Telecom	50	0.28	0.28	0.28	0.28	Telecom	6000	0.26	0.26	0.26	0.26
Utilities	55	0.19	0.19	0.18	0.18	Utilities	7000	0.19	0.19	0.19	0.19
Simple mean		0.27	0.27	0.26	0.22	Simple mean		0.27	0.27	0.26	0.24
<i>Difference</i>						<i>Difference</i>					
Energy	10	0.20	0.20	0.23	0.28	Oil & gas	500	0.18	0.18	0.19	0.22
Materials	15	0.07	0.07	0.09	0.04	Basic materials	1000	0.10	0.10	0.09	0.09
Industrials	20	0.10	0.11	0.14	0.15	Industrials	2000	0.08	0.08	0.13	0.17
Cons disc	25	0.05	0.08	0.09	0.10	Consumer services	5000	0.04	0.06	0.07	0.09
Cons stap	30	0.13	0.15	0.16	0.14	Consumer goods	3000	0.03	0.05	0.05	0.09
Health care	35	0.07	0.06	0.13	0.12	Health care	4000	0.07	0.07	0.09	0.14
Financials	40	0.07	0.11	0.12	0.13	Financials	8000	0.07	0.12	0.11	0.10
Info tech	45	0.09	0.10	0.09	0.11	Technology	9000	0.10	0.10	0.10	0.10
Telecom	50	0.32	0.32	0.32	0.32	Telecom	6000	0.09	0.09	0.09	0.09
Utilities	55	0.29	0.29	0.29	0.29	Utilities	7000	0.29	0.29	0.30	0.31
Simple mean		0.14	0.15	0.17	0.17	Simple mean		0.10	0.11	0.12	0.14
<i>t</i> -statistic		4.58***	5.23***	6.14***	5.66***	<i>t</i> -statistic		4.61***	5.69***	6.04***	6.54***

Abbreviations: GICS, Global Industry Classification System; ICB, Industry Classification Benchmark.
Significant at 10 per cent *, 5 per cent **, 1 per cent ***.

the total correlation difference between the two sectoring frameworks for each of the four levels. This difference is primarily generated by the differences existing for the inside-sector correlations.

Second, it is necessary to look at each of the sub-samples individually. In the first sub-sample between 1988 and 1992, inside-sector correlation ranges between 0.42 and 0.39 for the GICS framework, and between 0.37 and 0.40 for the ICB framework. In both cases, the values increase

between the 2- and 6-digit levels, and decrease between the 6- and 8 digit levels. The consequence is that the ICB system never reaches the level of inside-sector correlation reached by the GICS framework. Principally responsible for this higher correlation are only a few sectors. First, the Energy sector in the GICS framework reaches inside-sector correlation of 0.39 with 6-digit codes. ICB only reaches 0.34 at this stage and increases to 0.37 with 8-digit codes. In general, the Energy sector is a

Table 2: Inside versus outside correlation analysis for the two considered classification systems (period considered is 1993–1997)

<i>GICS sector</i>	<i>GICS code</i>	<i>GICS 2</i>	<i>GICS 4</i>	<i>GICS 6</i>	<i>GICS 8</i>	<i>ICB sector</i>	<i>ICB code</i>	<i>ICB 2</i>	<i>ICB 4</i>	<i>ICB 6</i>	<i>ICB 8</i>
<i>Within sector</i>						<i>Within sector</i>					
Energy	10	0.35	0.35	0.39	0.44	Oil & gas	500	0.33	0.33	0.36	0.38
Materials	15	0.26	0.26	0.35	0.22	Basic materials	1000	0.27	0.29	0.26	0.25
Industrials	20	0.26	0.27	0.28	0.29	Industrials	2000	0.23	0.23	0.25	0.27
Cons disc	25	0.18	0.21	0.23	0.23	Consumer services	5000	0.20	0.22	0.24	0.32
Cons stap	30	0.26	0.28	0.32	0.31	Consumer goods	3000	0.19	0.22	0.27	0.27
Health care	35	0.21	0.25	0.29	0.27	Health care	4000	0.26	0.26	0.29	0.27
Financials	40	0.27	0.27	0.22	0.22	Financials	8000	0.33	0.36	0.38	0.24
Info tech	45	0.54	0.54	0.55	0.55	Technology	9000	0.25	0.25	0.26	0.53
Telecom	50	0.34	0.38	0.38	0.36	Telecom	6000	0.26	0.26	0.24	0.36
Utilities	55	0.23	0.29	0.31	0.30	Utilities	7000	0.52	0.52	0.52	0.32
Simple mean		0.29	0.31	0.33	0.32	Simple mean		0.28	0.29	0.31	0.32
<i>Outside sector</i>						<i>Outside sector</i>					
Energy	10	0.11	0.11	0.11	0.10	Oil & gas	500	0.11	0.11	0.11	0.11
Materials	15	0.18	0.18	0.17	0.12	Basic materials	1000	0.18	0.18	0.17	0.14
Industrials	20	0.19	0.19	0.17	0.16	Industrials	2000	0.18	0.18	0.18	0.16
Cons disc	25	0.15	0.15	0.14	0.11	Consumer services	5000	0.14	0.14	0.14	0.16
Cons stap	30	0.16	0.15	0.16	0.14	Consumer goods	3000	0.16	0.16	0.16	0.13
Health care	35	0.15	0.16	0.15	0.12	Health care	4000	0.16	0.16	0.16	0.13
Financials	40	0.15	0.18	0.12	0.12	Financials	8000	0.18	0.18	0.18	0.13
Info tech	45	0.12	0.13	0.12	0.12	Technology	9000	0.13	0.13	0.13	0.12
Telecom	50	0.18	0.15	0.18	0.17	Telecom	6000	0.16	0.16	0.13	0.17
Utilities	55	0.13	0.12	0.12	0.11	Utilities	7000	0.12	0.12	0.12	0.13
Simple mean		0.15	0.15	0.14	0.13	Simple mean		0.15	0.15	0.15	0.14
<i>Difference</i>						<i>Difference</i>					
Energy	10	0.24	0.24	0.28	0.33	Oil & gas	500	0.22	0.22	0.24	0.27
Materials	15	0.09	0.09	0.18	0.10	Basic materials	1000	0.10	0.12	0.10	0.11
Industrials	20	0.07	0.08	0.11	0.13	Industrials	2000	0.06	0.06	0.08	0.10
Cons disc	25	0.03	0.06	0.09	0.12	Consumer services	5000	0.06	0.08	0.10	0.16
Cons stap	30	0.10	0.12	0.17	0.17	Consumer goods	3000	0.04	0.06	0.08	0.18
Health care	35	0.06	0.10	0.14	0.15	Health care	4000	0.10	0.10	0.13	0.14
Financials	40	0.12	0.09	0.11	0.11	Financials	8000	0.15	0.17	0.19	0.11
Info tech	45	0.42	0.42	0.44	0.44	Technology	9000	0.12	0.12	0.13	0.41
Telecom	50	0.15	0.23	0.20	0.19	Telecom	6000	0.11	0.11	0.11	0.19
Utilities	55	0.10	0.17	0.19	0.20	Utilities	7000	0.39	0.39	0.40	0.19
Simple mean		0.14	0.16	0.19	0.19	Simple mean		0.13	0.14	0.16	0.19
<i>t</i> -statistic		3.76***	4.65***	5.87***	5.54***			4.61***	5.69***	6.04***	6.54***

Abbreviations: GICS, Global Industry Classification System; ICB, Industry Classification Benchmark.
Significant at 10 per cent *, 5 per cent **, 1 per cent ***.

sector that shows high correlation. The ICB methodology broke this sector into two parts putting mining companies into the Basic Materials sector. The decision in itself is a reasonable one. However, stock returns of these mining companies seem to be influenced by the other commodity stocks, oil and gas stocks. At the same time, the Materials sector in the GICS framework is less homogeneous than the equivalent for the ICB system. The decision to classify the mining companies in either one of these

classes is therefore a defensible choice. Similar differences can be found for the Consumer Staples/Consumer Goods or the Consumer Discretionary/Consumer Services classes. Again, a difference in methodology leads to differences in inside-sector correlation. The challenge here is the comparability of the considered classes. Considering all four sub-sample periods, the inside-sector correlation of either one of the Consumer classes of the GICS system is higher than both the ICB classes. At the

Table 3: Inside versus outside correlation analysis for the two considered classification systems (period considered is 1998–2002)

<i>GICS Sector</i>	<i>GICS Code</i>	<i>GICS 2</i>	<i>GICS 4</i>	<i>GICS 6</i>	<i>GICS 8</i>	<i>ICB Sector</i>	<i>ICB Code</i>	<i>ICB 2</i>	<i>ICB 4</i>	<i>ICB 6</i>	<i>ICB 8</i>
<i>Within sector</i>						<i>Within sector</i>					
Energy	10	0.41	0.41	0.46	0.52	Oil & gas	500	0.42	0.42	0.46	0.46
Materials	15	0.37	0.37	0.40	0.28	Basic materials	1000	0.37	0.38	0.36	0.31
Industrials	20	0.31	0.32	0.31	0.30	Industrials	2000	0.29	0.30	0.32	0.31
Cons disc	25	0.26	0.29	0.29	0.27	Consumer services	5000	0.21	0.21	0.25	0.25
Cons stap	30	0.16	0.18	0.24	0.24	Consumer goods	3000	0.20	0.26	0.28	0.28
Health care	35	0.22	0.26	0.30	0.31	Health care	4000	0.15	0.15	0.17	0.24
Financials	40	0.37	0.37	0.33	0.33	Financials	8000	0.32	0.36	0.39	0.40
Info tech	45	0.33	0.33	0.35	0.35	Technology	9000	0.39	0.39	0.40	0.41
Telecom	50	0.32	0.42	0.42	0.42	Telecom	6000	0.18	0.20	0.18	0.18
Utilities	55	0.37	0.42	0.42	0.44	Utilities	7000	0.34	0.34	0.34	0.36
Simple mean		0.31	0.34	0.35	0.35	Simple mean		0.29	0.30	0.32	0.32
<i>Outside sector</i>						<i>Outside sector</i>					
Energy	10	0.20	0.20	0.20	0.19	Oil & gas	500	0.21	0.21	0.21	0.20
Materials	15	0.24	0.24	0.23	0.16	Basic materials	1000	0.24	0.24	0.23	0.19
Industrials	20	0.22	0.22	0.21	0.20	Industrials	2000	0.23	0.23	0.23	0.21
Cons disc	25	0.20	0.20	0.20	0.16	Consumer services	5000	0.19	0.18	0.18	0.17
Cons stap	30	0.13	0.13	0.13	0.13	Consumer goods	3000	0.19	0.19	0.19	0.15
Health care	35	0.16	0.16	0.16	0.14	Health care	4000	0.14	0.14	0.14	0.14
Financials	40	0.17	0.17	0.14	0.14	Financials	8000	0.21	0.21	0.21	0.21
Info tech	45	0.15	0.15	0.15	0.15	Technology	9000	0.16	0.16	0.16	0.15
Telecom	50	0.21	0.21	0.21	0.19	Telecom	6000	0.17	0.14	0.13	0.13
Utilities	55	0.16	0.16	0.15	0.15	Utilities	7000	0.16	0.16	0.16	0.16
Simple mean		0.18	0.18	0.18	0.16	Simple mean		0.19	0.18	0.18	0.17
<i>Difference</i>						<i>Difference</i>					
Energy	10	0.21	0.21	0.26	0.33	Oil & gas	500	0.21	0.21	0.25	0.27
Materials	15	0.13	0.13	0.17	0.12	Basic materials	1000	0.13	0.14	0.13	0.12
Industrials	20	0.08	0.10	0.10	0.10	Industrials	2000	0.07	0.07	0.09	0.10
Cons disc	25	0.06	0.09	0.10	0.10	Consumer services	5000	0.03	0.03	0.07	0.08
Cons stap	30	0.03	0.05	0.10	0.11	Consumer goods	3000	0.03	0.03	0.07	0.10
Health care	35	0.06	0.10	0.14	0.17	Health care	4000	0.01	0.01	0.03	0.10
Financials	40	0.20	0.20	0.19	0.19	Financials	8000	0.11	0.15	0.19	0.19
Info tech	45	0.19	0.19	0.21	0.21	Technology	9000	0.24	0.24	0.25	0.26
Telecom	50	0.12	0.21	0.21	0.23	Telecom	6000	0.01	0.05	0.05	0.05
Utilities	55	0.21	0.25	0.27	0.29	Utilities	7000	0.19	0.19	0.19	0.20
Simple mean		0.13	0.15	0.18	0.18	Simple mean		0.10	0.11	0.13	0.15
<i>t-statistic</i>		5.78***	7.31***	8.68***	7.21***			3.63***	4.60***	5.37***	6.46***

Abbreviations: GICS, Global Industry Classification System; ICB, Industry Classification Benchmark.
Significant at 10 per cent *, 5 per cent **, 1 per cent ***.

same time, the other GICS class performs average, which can, in such a case, be expected. For the 1998–2002 sub-sample and the 2003–2007 sub-sample, the Consumer Staples category correlation was low in general terms across all levels. This is of course interesting as this second period is considered to be, at first, a difficult one with the tech-bubble bursting, which should have promoted non-discretionary spending among consumers, and later the inflation of consumption with the economic revival between 2003 and 2007. It could suggest that

the companies inside the Consumer Staples sector do not really perform as a sector but rather as individual companies. The question then arises whether the segmentation into discretionary and non-discretionary items is a good one. The same point can, however, be made for the ICB methodology, as it resembles more to an average of the effects seen in the GICS system.

The last sector to be considered as a particular case is the Telecoms sector. In the GICS framework, this sector enjoys a significantly higher inside-sector correlation

Table 4: Inside versus outside correlation analysis for the two considered classification systems (period considered is 2003–2007)

<i>GICS sector</i>	<i>GICS code</i>	<i>GICS 2</i>	<i>GICS 4</i>	<i>GICS 6</i>	<i>GICS 8</i>	<i>ICB sector</i>	<i>ICB code</i>	<i>ICB 2</i>	<i>ICB 4</i>	<i>ICB 6</i>	<i>ICB 8</i>
<i>Within sector</i>						<i>Within sector</i>					
Energy	10	0.52	0.52	0.55	0.58	Oil & gas	500	0.47	0.47	0.50	0.49
Materials	15	0.28	0.28	0.31	0.27	Basic materials	1000	0.25	0.28	0.28	0.28
Industrials	20	0.31	0.35	0.34	0.31	Industrials	2000	0.27	0.27	0.31	0.32
Cons disc	25	0.27	0.31	0.33	0.31	Consumer services	5000	0.23	0.25	0.27	0.27
Cons stap	30	0.13	0.15	0.17	0.19	Consumer goods	3000	0.16	0.20	0.23	0.22
Health care	35	0.14	0.17	0.18	0.18	Health care	4000	0.12	0.12	0.18	0.14
Financials	40	0.30	0.30	0.29	0.29	Financials	8000	0.31	0.37	0.40	0.39
Info tech	45	0.40	0.40	0.40	0.40	Technology	9000	0.28	0.28	0.32	0.30
Telecom	50	0.33	0.41	0.43	0.43	Telecom	6000	0.28	0.27	0.23	0.23
Utilities	55	0.27	0.31	0.32	0.31	Utilities	7000	0.35	0.35	0.34	0.34
Simple mean		0.30	0.32	0.33	0.33	Simple mean		0.27	0.29	0.31	0.30
<i>Outside sector</i>						<i>Outside sector</i>					
Energy	10	0.09	0.09	0.09	0.09	Oil & gas	500	0.10	0.10	0.10	0.10
Materials	15	0.18	0.18	0.17	0.14	Basic materials	1000	0.17	0.17	0.15	0.17
Industrials	20	0.18	0.18	0.17	0.16	Industrials	2000	0.17	0.17	0.16	0.17
Cons disc	25	0.18	0.18	0.18	0.15	Consumer services	5000	0.17	0.17	0.16	0.17
Cons stap	30	0.10	0.10	0.10	0.10	Consumer goods	3000	0.15	0.15	0.13	0.15
Health care	35	0.11	0.11	0.11	0.11	Health care	4000	0.10	0.10	0.10	0.10
Financials	40	0.19	0.19	0.19	0.19	Financials	8000	0.18	0.18	0.18	0.18
Info tech	45	0.12	0.12	0.12	0.12	Technology	9000	0.18	0.18	0.18	0.18
Telecom	50	0.18	0.18	0.18	0.17	Telecom	6000	0.21	0.18	0.15	0.15
Utilities	55	0.17	0.17	0.17	0.16	Utilities	7000	0.12	0.12	0.12	0.12
Simple mean		0.15	0.15	0.15	0.14	Simple mean		0.16	0.15	0.14	0.15
<i>Difference</i>						<i>Difference</i>					
Energy	10	0.43	0.43	0.45	0.49	Oil & gas	500	0.37	0.37	0.40	0.39
Materials	15	0.10	0.10	0.14	0.13	Basic materials	1000	0.08	0.10	0.13	0.11
Industrials	20	0.13	0.17	0.17	0.15	Industrials	2000	0.09	0.09	0.14	0.14
Cons disc	25	0.09	0.13	0.15	0.15	Consumer services	5000	0.06	0.09	0.10	0.10
Cons stap	30	0.03	0.05	0.07	0.09	Consumer goods	3000	0.08	0.10	0.14	0.12
Health care	35	0.03	0.06	0.06	0.08	Health care	4000	0.02	0.02	0.08	0.03
Financials	40	0.12	0.12	0.10	0.10	Financials	8000	0.13	0.19	0.22	0.21
Info tech	45	0.29	0.29	0.28	0.28	Technology	9000	0.10	0.10	0.14	0.12
Telecom	50	0.15	0.24	0.25	0.26	Telecom	6000	0.07	0.08	0.07	0.07
Utilities	55	0.10	0.14	0.15	0.15	Utilities	7000	0.23	0.23	0.22	0.22
Simple mean		0.15	0.17	0.18	0.19	Simple mean		0.12	0.14	0.16	0.15
<i>t</i> -statistic		3.82***	4.69***	4.97***	4.71***			3.48***	4.08***	5.18***	4.46***

Abbreviations: GICS, Global Industry Classification System; ICB, Industry Classification Benchmark.
Significant at 10 per cent *, 5 per cent **, 1 per cent ***.

across all sub-samples and all levels of the sectoring systems. The approach used by the GICS framework seems clearly better as it generates a difference between inside and outside correlation ranging between 0.12 and 0.32. Other sectors, such as the Information Technology sector or the Financial sector, do not present significant and persistent differences between the two systems.

The general conclusion, in terms of the difference between inside and outside correlation, clearly suggests the GICS

systems to be of higher quality. The difference is mostly because of three of the main sectors showing higher inside correlations. As the outside correlations are remarkably similar, the overall correlation difference remains in favor of the GICS system. Consequently, a higher percentage of the return volatility of a stock is accounted for by its industry peers in the case of the GICS system. For example, 16 per cent of the Information Technology stocks' return volatility can be accounted for by its industry peers, whereas only 8 per cent, in the case of

the ICB system, for the digit level. From this viewpoint, the diversification one could achieve using a 2-digit GICS code is approximately similar to the level of diversification of a 4-digit ICB system.

The obvious question that now remains concerns the consequences of this higher inside-sector correlation in terms of the average return of the sectors, the beta of the sectors compared to a market benchmark and the possible correlation shifts one can expect in the case of market movements. All three of these questions need to be answered. The proposed solution consists of estimating both an SIMM and a QMM of the 2-digit sectors for each of the classification systems. As the previous analysis has already indicated, a marked difference exists even at this 10-sector level.

Both models are estimated for a 20-year sample, between 1988 and 2007 included. All companies present in the total sample are included. Table 5 presents the results for the SIMM. The first observation is the high significance of the model. All estimated parameters are significant up to 1 per cent and the adjusted R^2 is around 70 per cent. As the outside-sector correlation remains similar for both systems, one would indeed expect that the beta values for the two systems would give a similar picture. Alpha values, on the contrary, could vary as they would then reflect a higher homogeneity inside the sectors. The results of the analysis give a different picture. The excess returns of the two systems compared to the considered benchmark, the S&P 500 index over the total period, are very similar. The higher inside-sector correlation for a number of sectors, in the GICS systems, does not seem to translate into a marked difference in excess returns. The outside-sector correlation was indeed similar for both systems. The higher degree of homogeneity provided by using the GICS framework, therefore, implies that the return volatility of the considered sector is explained to a higher degree by its industry peers.

The beta values also present interesting features. On the one hand, the very similar level of outside-sector correlation for both sectoring systems leads one to believe that both systems will also behave similarly with respect to the market benchmark. On the other hand, the higher inside correlation for GICS should result in lower beta values as industry peers are more responsible for the return volatility. Four out of 10 sectors show significantly lower beta values for the GICS framework. The remaining six beta values are relatively similar even though the GICS values are higher than those found for the ICB framework. The primary differences are found between those sectors that show methodology differences between the two systems. The Energy and Materials sectors are among those showing major methodology differences. The GICS system shows a constant relatively higher inside correlation for these sectors; however, the outside-sector correlation remains relatively similar for the two systems. This implies that a greater proportion of the return volatility for these sectors is explained by the sector itself rather than by the market. This effect is visible in the values for beta. Consequently, the Materials sector, which tends to show similar or higher inside-sector correlation for the ICB framework, also has a lower beta value. This result lets one believe that the grouping choices of the Energy or Materials sectors are equivalent.

More interesting is the fact that the Consumer Discretionary sector has a lower beta value of 0.859 than the Consumer Services sector. It is true that the two are not really comparable; however, the beta is approximately similar to the one in the Consumer Goods sector. This value seems to indicate that stock returns of the Consumer Discretionary sector is less dependent on economic cycles than Non-Discretionary stock returns.

All other sectors show higher beta values for the GICS framework. The values do not differ greatly from those in the ICB system;

Table 5: Regression results single index market model for the two systems over the period 1988–2007

GICS sector	GICS				ICB						
	GICS code	α	β	$adjR^2$	ICB sector	ICB code	α	β	$adjR^2$		
Energy	10	Beta t-statistic P-value	-0.010*** -6.196 0.000	1.011*** 25.107 0.000	0.720 — —	Oil & gas	500	Beta t-statistic P-value	-0.009*** -5.289 0.000	1.115*** 25.808 0.000	0.731 — —
Materials	15	Beta t-statistic P-value	-0.011*** -7.521 0.000	0.746*** 19.256 0.000	0.601 — —	Basic materials	1000	Beta t-statistic P-value	-0.010*** -6.502 0.000	0.839*** 22.252 0.000	0.669 — —
Industrials	20	Beta t-statistic P-value	-0.012*** -7.746 0.000	0.952*** 25.208 0.000	0.721 — —	Industrials	2000	Beta t-statistic P-value	-0.011*** -7.797 0.000	0.919*** 25.820 0.000	0.731 — —
Cons disc	25	Beta t-statistic P-value	-0.011*** -9.656 0.000	0.862*** 29.823 0.000	0.784 — —	Consumer services	5000	Beta t-statistic P-value	-0.010*** -9.883 0.000	0.904*** 34.952 0.000	0.833 — —
Cons stap	30	Beta t-statistic P-value	-0.010*** -7.543 0.000	0.901*** 26.179 0.000	0.736 — —	Consumer goods	3000	Beta t-statistic P-value	-0.010*** -7.358 0.000	0.859*** 25.368 0.000	0.724 — —
Health care	35	Beta t-statistic P-value	-0.011*** -6.727 0.000	0.986*** 24.318 0.000	0.707 — —	Health care	4000	Beta t-statistic P-value	-0.010*** -6.381 0.000	0.962*** 23.259 0.000	0.688 — —
Financials	40	Beta t-statistic P-value	-0.010*** -8.035 0.000	0.889*** 29.376 0.000	0.779 — —	Financials	8000	Beta t-statistic P-value	-0.010*** -7.897 0.000	0.877*** 28.466 0.000	0.768 — —
Information technology	45	Beta t-statistic P-value	-0.010*** -7.458 0.000	1.045*** 30.780 0.000	0.794 — —	Technology	9000	Beta t-statistic P-value	-0.009*** -6.543 0.000	1.030*** 29.933 0.000	0.785 — —
Telecommunication	50	Beta t-statistic P-value	-0.010*** -4.641 0.000	0.843*** 15.496 0.000	0.494 — —	Telecommunication	6000	Beta t-statistic P-value	-0.010*** -9.864 0.000	0.904*** 36.025 0.000	0.841 — —
Utilities	55	Beta t-statistic P-value	-0.011*** -7.988 0.000	0.960*** 27.436 0.000	0.754 — —	Utilities	7000	Beta t-statistic P-value	-0.010*** -7.403 0.000	0.908*** 25.639 0.000	0.728 — —

Abbreviations: GICS, Global Industry Classification System; ICB, Industry Classification Benchmark.
Significant at 10 per cent *, 5 per cent **, 1 per cent ***.

the fact that they are higher does suggest that the GICS system is a less independent form of market movement than the ICB system. The general conclusion found in the previous analysis therefore holds. The GICS systems' general quality hinges upon a number of very well-selected sectors, combined with average performance across the others. The overall higher cohesion, found when looking at the higher difference between inside- and outside-sector correlation, could therefore be misleading. The qualities of the GICS system lie in some of its sectors, whereas the ICB system has a more consistent performance across its sectors.

The last question that needs to be resolved is the stability of each of the systems in the event of market shifts. Put differently, what effect will a shift in the market returns have on the returns of each of the separate sectors. A QMM is estimated using the same sub-sample and the same S&P 500 index as market proxy. Table 6 shows the results for the estimated model. Using the estimated values and equations (9) and (10), the variations in the beta values of the different sectors can be computed for a variation in the market.

The results in Table 7 are very similar to those found in the estimation of the SIMM. This is an indication of the quality of the QMM estimations. The significance of the found quadratic terms is, however, less high than could be expected. Only a small number of sectors show a significant quadratic term. A number of conclusions can be drawn from this result.

First, the overall stability of the sectors with respect to the evolution of the S&P 500 index can be deduced from these findings. The relation between the returns of a sector and those of the index as a whole seems to be linear in most cases. Variations of the index do not generate shifts in the correlation between the index and the sector. It implies that for those sectors sharing insignificant quadratic terms for both sectoring systems, the concern for stability of the sector is limited, as far as large-cap companies are concerned.

Second, those classes that do have significant quadratic terms deserve particular attention. Both the GICS and the ICB system have related sectors with significant quadratic terms. Rather than telling something about the classification systems, it is an indication for the nature of these sectors. In that respect, it is interesting to note that these same sectors also presented interesting features in the preceding analysis. The delta beta column in Table 7 represents the expected change in beta between the benchmark and the sector in the event of a 1 per cent change in the benchmark's returns. The Energy sector, or the Oil and Gas sector shows for both frameworks, a negative change in beta and therefore a negative change in correlation. This would imply that both sectors show a decrease in correlation when the S&P 500 index drops. The correlation shift, however, is equivalent for both sectoring systems. This same conclusion can be drawn for the two other sectors for which a correlation shift is significant.

In general terms, it is therefore clear that the sectors are very similar in terms of stability to variations in the market benchmark. The initial higher homogeneity that was found for the GICS framework seems to be less present when market models are considered. The two models estimate the relation between the sectors for both systems and the evolution of the market benchmark. Neither of the tested systems seems to construct more homogeneous or stable sectors when the evolution with respect to the market is analyzed.

CONCLUSION

The article assesses two major sectoring systems for grouping equity into classes. The two considered systems are particular as they are rivals in terms of methodology, structure and intended users. Both are primarily aimed at practitioners, both are developed by index providers and both share a similar four-layered structure. The primary objective is

Table 6: Regression results for the two quadratic markets models for the two considered classification systems

GICS sector	GICS				ICB							
	GICS code	a	b1	b2	adjR ²	ICB sector	ICB Code	Beta	a	b1	b2	adjR ²
Energy	10	Beta	-0.007***	1.008***	-1.947***	Oil & gas	500	Beta	-0.005***	1.111***	-2.242***	0.7
		t-statistic	-3.551	25.495	-3.222			t-statistic	-2.656	26.288	-3.468	—
		P-value	0.000	0.000	0.001			P-value	0.008	0.000	0.001	—
Materials	15	Beta	-0.012***	0.746***	0.215	Basic materials	1000	Beta	-0.010***	0.839***	0.354	0.7
		t-statistic	-6.500	19.224	0.362			t-statistic	-5.785	22.232	0.613	—
		P-value	0.000	0.000	0.717			P-value	0.000	0.000	0.541	—
Industrials	20	Beta	-0.010***	0.951***	-0.842	Industrials	2000	Beta	-0.010***	0.919***	-0.439	0.7
		t-statistic	-5.724	25.222	-1.462			t-statistic	-6.105	25.772	-0.805	—
		P-value	0.000	0.000	0.145			P-value	0.000	0.000	0.422	—
Cons disc	25	Beta	-0.012***	0.863***	0.868**	Consumer goods	3000	Beta	-0.012***	0.906***	1.054***	0.8
		t-statistic	-9.229	30.041	1.977			t-statistic	-9.870	35.456	2.698	—
		P-value	0.000	0.000	0.049			P-value	0.000	0.000	0.007	—
Cons stap	30	Beta	-0.010***	0.901***	0.028	Consumer services	5000	Beta	-0.010***	0.859***	0.099	0.7
		t-statistic	-6.349	26.118	0.053			t-statistic	-6.270	25.315	0.192	—
		P-value	0.000	0.000	0.958			P-value	0.000	0.000	0.848	—
Health care	35	Beta	-0.010***	0.986***	-0.178	Health care	4000	Beta	-0.010***	0.962***	-0.225	0.7
		t-statistic	-5.482	24.257	-0.287			t-statistic	-5.155	23.201	-0.355	—
		P-value	0.000	0.000	0.775			P-value	0.000	0.000	0.723	—
Financials	40	Beta	-0.010***	0.889***	-0.008	Financials	8000	Beta	-0.010***	0.877***	0.027	0.8
		t-statistic	-6.722	29.305	-0.018			t-statistic	-6.647	28.400	0.056	—
		P-value	0.000	0.000	0.986			P-value	0.000	0.000	0.955	—
Information technology	45	Beta	-0.010***	1.045***	0.042	Technology	9000	Beta	-0.009***	1.030***	-0.113	0.8
		t-statistic	-6.293	30.709	0.081			t-statistic	-5.366	29.859	-0.215	—
		P-value	0.000	0.000	0.935			P-value	0.000	0.000	0.830	—
Telecommunication	50	Beta	-0.012***	0.845***	1.519*	Telecommunication	6000	Beta	-0.011***	0.905***	1.020***	0.8
		t-statistic	-4.912	15.612	1.834			t-statistic	-9.849	36.539	2.692	—
		P-value	0.000	0.000	0.068			P-value	0.000	0.000	0.008	—
Utilities	55	Beta	-0.011***	0.960***	0.266	Utilities	7000	Beta	-0.011***	0.908***	0.214	0.7
		t-statistic	-6.966	27.397	0.496			t-statistic	-6.419	25.596	0.395	—
		P-value	0.000	0.000	0.620			P-value	0.000	0.000	0.694	—

Abbreviations: GICS, Global Industry Classification System; ICB, Industry Classification Benchmark.
Significant at 10 per cent *, 5 per cent **, 1 per cent ***.

Table 7: Alpha, change in beta and change in correlation

	GICS				ICB					
	GICS sector	GICS code	α	$\Delta\beta$	$\Delta\rho$	ICB sector	ICB Code	α	$\Delta\beta$	$\Delta\rho$
Energy		10	-0.026***	-0.019***	-0.023***	Oil & gas	500	-0.028***	-0.022***	-0.029***
Materials		15	-0.01	0.002	0.002	Basic materials	1000	-0.007	0.004	0.004
Industrials		20	-0.019	-0.008	-0.009	Industrials	2000	-0.015	-0.004	-0.005
Consumer discretionary		25	-0.004**	0.009**	0.008**	Consumer goods	3000	-0.001***	0.011***	0.011***
Consumer staples		30	0.01	0.000	0.000	Consumer services	5000	-0.009	0.001	0.001
Health care		35	-0.012	-0.002	-0.002	Health care	4000	-0.012	-0.002	-0.002
Financials		40	-0.01	0.000	0.000	Financials	8000	-0.009	0.000	0.000
Information technology		45	-0.01	0.000	0.000	Technology	9000	-0.010	-0.001	-0.001
Telecommunication		50	0.0027*	0.015*	0.018*	Telecommunication	6000	-0.001***	0.010***	0.010***
Utilities		55	-0.009	0.003	0.003	Utilities	7000	-0.009	0.002	0.002

Abbreviations: GICS, Global Industry Classification System; ICB, Industry Classification Benchmark.

Significant at 10 per cent *, 5 per cent **, 1 per cent ***.

therefore to compare the two frameworks. Contrary to prior research, however, this article focuses on exact behavior of the sectors as defined by each of the systems. This question was split into three parts. First, what is the actual homogeneity of each sector for all layers of the system across a long period of time. The second question that immediately follows concerns the actual behavior of these sectors in comparison with the evolution of a benchmark: the found differences in the first analysis reflected in the returns and betas of the sectors. The third question concerns the fact of stability with respect to shifts in the returns of the market benchmark. The first analysis does indeed show considerable differences between the two considered frameworks. The GICS framework has consistently higher inside-sector correlations, meaning that a larger proportion of the return volatility of a stock is explained by companies belonging to the same sector. The higher inside-sector correlation is because of a limited number of sectors for which the two considered frameworks have different methodologies. The outside-sector correlation, however, remains stable and similar for both frameworks. In order to test whether the higher inside sector correlation has a parallel in the performance of a sector, an SIMM is used. The results again are interesting. The betas for the GICS system are lower for those sectors showing a consistently higher inside-sector correlation. For all other sectors, the GICS betas are higher. These results bring us to the conclusion that at least a part of the quality of the GICS framework is because of the choices made for only a limited number of sectors. These sectors are more homogeneous than their counterparts in the ICB framework. This conclusion does not give us sufficient evidence that either one of the systems outperforms the other. Both seem to have strengths and weaknesses, but neither is sufficiently convincing to be hailed superior. Using a QMM, the last part of the test is performed. Neither of the two systems

seems considerably vulnerable to market shocks. All stocks are affected equally by shocks and will not generate major differences in their correlation with respect to market movement. The principal conclusion of this article is therefore twofold. On the one hand, using simple correlation analysis, the GICS framework seems to outperform its rival. Looking, however, in detail at the results, the better results for GICS are based on only a few technical decisions that differ from ICB. Using market models, these differences persist but become less important as the overall picture is less in favor of GICS. The beta values show, in more detail, that some sectors clearly outperform those in the ICB system, but all others tend to do slightly worse. The results should be read with interest. However, tests on much larger samples with both small- and large-cap stocks could yield further interesting conclusions. We believe that those tests will yield similar, if not more pronounced differences. This would lead to the conclusion that GICS is better for some sectors, whereas ICB remains more consistent across all sectors. Both systems should therefore be considered globally similar but locally different.

NOTE

1. The data used for this study were provided by Fortis Bank – Merchant Banking – Dr D.-E. Baestaens.

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